Michael Cogan

Software Developer & Engineering Technologist

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SUMMARY

Software developer who builds working systems from the ground up — websites, data pipelines, AI assistants, and hardware that senses its environment. Comfortable across the full stack (from the browser to the server to the chip) and known for explaining complex things clearly and collaborating well with non-technical stakeholders.

SKILLS

Programming: Python, C, C++, JavaScript, SQL, and web fundamentals (HTML/CSS)

Databases & hosting: PostgreSQL and SQLite (relational data), plus cloud deployment on modern platforms

Hardware & systems: Raspberry Pi and single-board computers, sensor and camera integration, Linux command line

AI & data: Local language models (Ollama), vector search (ChromaDB), NumPy/SciPy, Pandas, Matplotlib

People skills: Clear communication, teaching and mentoring, teamwork, adaptability, careful attention to detail

CERTIFICATIONS

Intro to Software Engineering — IBM / Coursera

CI/CD for Software Developers — Codio / Coursera

Software Engineer — HackerRank

Python (Basic) — HackerRank

EDUCATION

B.S. Electrical & Computer Engineering Technology — Expected 2027

Valencia College — Orlando, FL

Associate in Arts, Physics — Graduated 2025

Valencia College — Orlando, FL

EXPERIENCE

Freelance Web Developer — Sep 2026 — Present

Front Line Refinement — remote (paid contract)

- Worked one-on-one with a non-technical business owner, translating their goals into a live website they can update themselves through a content-management system.
- Built the backend in Node.js (JavaScript on the server) with a PostgreSQL database, plus online waivers and an appointment-booking calendar with scheduling conflict detection.
- Secured customer data with password hashing, login rate-limiting, and cross-site request protections, then deployed it to the cloud.

Assistant Researcher — 2025 — 2026

Rings Research Group — Orlando, FL

- Processed large NASA astronomy datasets using Python (with NumPy and Matplotlib), turning raw measurements into clear charts and summaries for the research team.
- Built a repeatable, documented data-processing workflow that saved the team time on later projects.

HIGHLIGHTED PROJECTS

Eigen — A Personal AI Assistant

- Built an AI assistant that runs entirely on a home server (no cloud), using a local language model and a vector database so it can "remember" past conversations.
- Connected a Raspberry Pi with a camera and a distance sensor so the assistant can see and sense its surroundings, and made it safe to share publicly.

Atom Wavefunction Visualizer

- Created a real-time 3D visualization of an atom's electron cloud in C++ with OpenGL shaders, rendering the shape directly on the graphics card.

Interactive Physics Simulator

- Built an interactive webpage that simulates springs and oscillating systems using Python (compiled to run in the browser via WebAssembly) and SciPy.

OTHER EXPERIENCE

Physics & Mathematics Tutor Ongoing

Educate 2 Elevate — Orlando, FL

- Explained difficult math and physics concepts in plain language, tailoring each lesson to the student's pace and learning style.
- Patiently adapted sessions for students managing anxiety, helping build confidence alongside technical skill.

Infantry Section Leader 2018 — 2024

Army National Guard — River Falls, WI

- Managed electronic and network equipment to keep everything mission-ready.
- Led teams under pressure, including during COVID-19 relief missions.
- Trained groups of up to 100 people, developing strong communication and leadership skills.